

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459926.6529 +/- 0.0071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.034 +/- 0.062
Transit depth (Rp/Rs)^2	0.12 +/- 0.42 [%]
Orbital Inclination (inc)	81.86 +/- 2.85
Airmass coefficient 1 (a1)	1.04 +/- 0.48
Airmass coefficient 2 (a2)	-0.1 +/- 0.32
Scatter in the residuals of the lightcurve fit is	42.64 %
Best Comparison Star	#2 - [57, 39]
Optimal Aperture	2.77
Optimal Annulus	3.01
Transit Duration (day)	0.025 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.034 ± 0.062 vs 0.1646 (deviation 2.11σ)
Duration fit vs published	0.025 d vs 0.0758 d (deviation 1.82σ)
Mid-time O–C	-16.9 min
Depth SNR	0.5
Residual scatter	42.64 %

Light curve

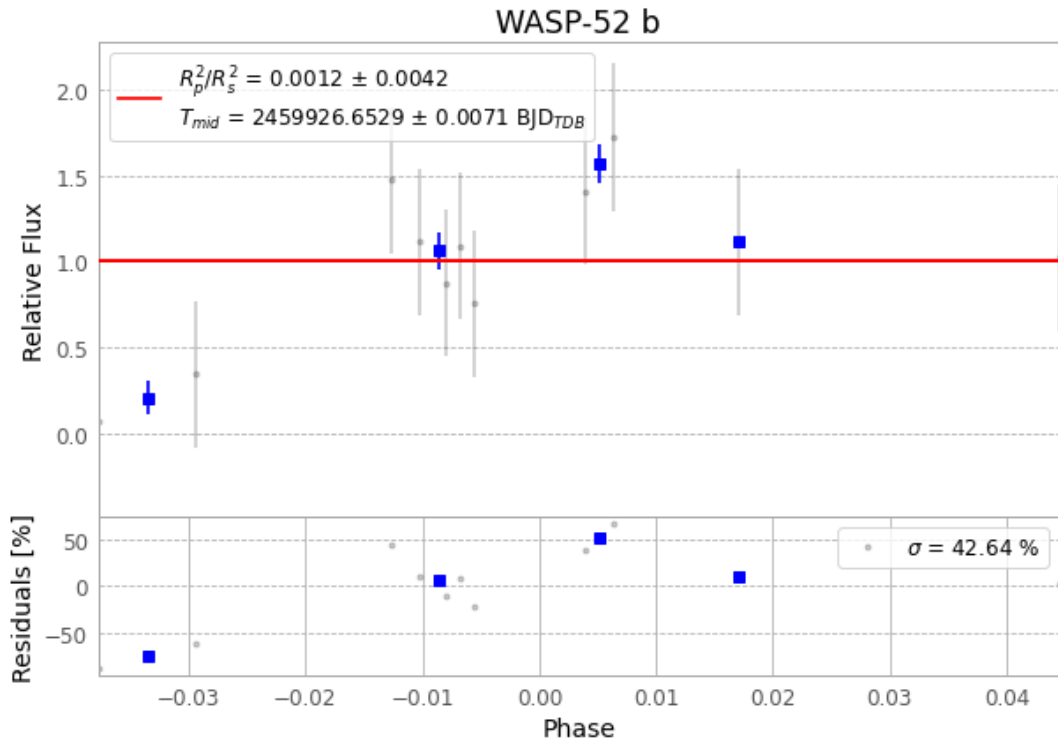


Figure 1 — FinalLightCurve_WASP-52 b_2022-12-12.png (SHA-256 b783a368a17c7363...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [94, 403], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1bc42cf8de946dce...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

77 FITS frames (51 MB), WASP-52221213020316.FITS → WASP-52221213055115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-221213.log (c476a40af990...), FinalLightCurve_WASP-52 b_2022-12-12.png (b783a368a17c...), FinalLightCurve_WASP-52 b_2022-12-12.csv (025bc1fbbe2f...)

Compute resources

Wall time 105.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 17:47Z.